

MACHINE LEARNING With TensorFlow

PRIYA RAVINDRAN

The Google Brain team developed a software library to perform machine learning across a range of tasks. Their aim was to cater to the needs of their machine learning systems, those that were capable of building and training neural networks. The software was meant to help such systems detect and decipher patterns and correlations, just like the way human beings learn and reason.

In November 2015, Google released this tool under Apache 2.0 licence, making it open to use and providing everyone an opportunity to work on their own artificial intelligence (AI) based projects. By June 2016, 1500 repositories on GitHub mentioned the software, of which only five were from Google.

The tool under discussion is TensorFlow. What is it about this tool that makes it suitable for machine learning applications? How do you use the tool? Why does Google hold it at such high regard, and why are so many people contributing to it? This article gives you an overview of how TensorFlow flows.

Working with TensorFlow

When you import TensorFlow into Python environment, you get complete access over its classes, methods and symbols. Take TensorFlow operations and arrange these into a graph of nodes called the computational graph. Typically, each node takes tensors as inputs and produces a corresponding output tensor. Values for the nodes get evaluated as and when a session is run. Com-

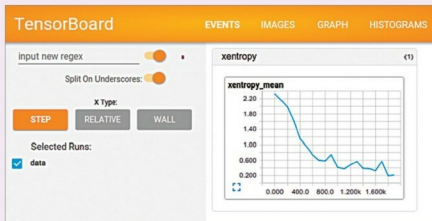


Fig. 1: A fully-configured TensorBoard

About TensorFlow

Developed by: Google Brain Team
Last stable release: 1.0
Held on: GitHub Repository
Written in: Python and C++
Supports: Linux, Mac and Windows
Under: Apache open source licence
Can be found at: www.tensorflow.org

bine nodes with operations—also nodes in a certain form—to build more complicated computations.

Customise and improvise. To tune this for your machine learning application, you need to construct the model such that it can take arbitrary inputs and deliver outputs accordingly. The way to do this with TensorFlow is to add variables, reflecting trainable parameters. Each of these has a type and an initial value, letting you tune your system to required behaviour.

How do you know if your system is functioning exactly the way you intended it to? Simple, just introduce a loss function. TensorFlow provides optimisers that slowly change each

variable so that loss function can be minimised. There are also higher abstractions for common patterns, structures and functionality.

Multiple APIs for easier control

As a new user to any software, it is important to enjoy the experience. TensorFlow is built with that mindset, with the highest-level application program interface (API) tuned for easy learning and usage. With experience, you will learn how to handle the tool, and what modification will result in what kind of change to the entire functionality.

It is then obvious to want to be able to work around the model and have fine levels of control over the same. TensorFlow core API, the lowest-level API, helps you achieve this fine control. Other higher-level APIs are built on top of this very core. Higher the level of the API, the easier it is to perform repetitive tasks and to keep the flow consistent between multiple users.